

Systems Engineering Challenges on Mars Reconnaissance Orbiter Mission

Glen Havens¹

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, 91109

Mars Reconnaissance Orbiter (MRO) began its journey to Mars on August 12, 2005 with a spectacular launch on an Atlas V launch vehicle. As with any deep space mission, the journey for MRO engineers began many years earlier with the design and development of the MRO spacecraft and its plan for operations. This is the period in the mission's evolution from concept to reality, where high-level requirements transform into an integrated system ready for flight. Even after launch, final development of the primary science phase operations continued for another full year as MRO completed a 7-month cruise, Mars orbit insertion, and a 5-month aerobraking phase to place the spacecraft in its final science orbit. In order for MRO to fulfill its destiny to dramatically advance our knowledge and understanding of Mars, the systems engineering team had to first overcome some key challenges beyond previous Mars missions. Examples of these key challenges include: 1) development of an integrated science planning process, utilizing spacecraft-loaded Integrated Target List (ITL), to support six science instruments with diverse scientific objectives; 2) implementation of an end-to-end data system to support downlink data rates up to 6 Mbit/sec, and deliver data from deep space network antennas, to JPL and distribute data products to science teams; 3) use of the spacecraft's Solid State Recorder (SSR) to store science data, coupled with a ground commanded data retransmission strategy to optimize data return; and 4) design of Operational Readiness Tests (ORTs) utilizing high fidelity spacecraft test-beds to validate operational scenarios, processes and flight team certification. Using these examples, this paper will explore the corresponding driving requirements, the technical obstacles encountered, and the corresponding system engineering responses that have successfully enabled MRO to begin its primary science mission on November 11, 2006.

I. Introduction

COMPLEX projects, such as deep space missions, are a system of systems requiring a specialized engineering approach known as system engineering to architect, design, integrate, test, and operate these systems at each level. The Mars Reconnaissance Orbiter (MRO) project is an example of such a system of systems. At the highest level, it contains the flight and ground systems linked together by a mission design. The flight system can be decomposed into the MRO spacecraft, with multiple subsystems and payloads including six scientific instruments, and the launch vehicle, which plays a brief but important role, lifting MRO away from earth and into its trajectory toward Mars. Likewise the ground system, or Mission Operations System (MOS), can be decomposed into the mission operations teams, mission operations processes, and the software, tools, and facilities, known collectively as the ground data system (GDS), which includes the Deep Space Network (DSN) of antennas.

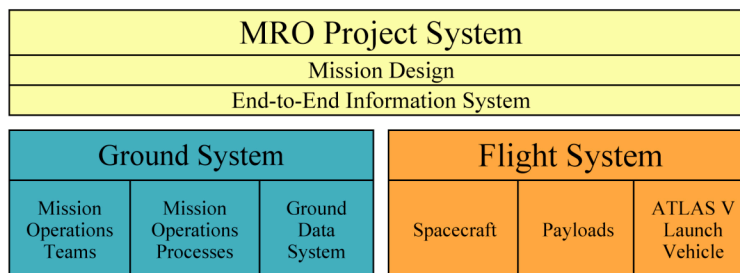


Figure 1. MRO system of systems

¹ Mission Operations Systems Engineer, MRO Project, 4800 Oak Grove Drive, Mailstop 264-535

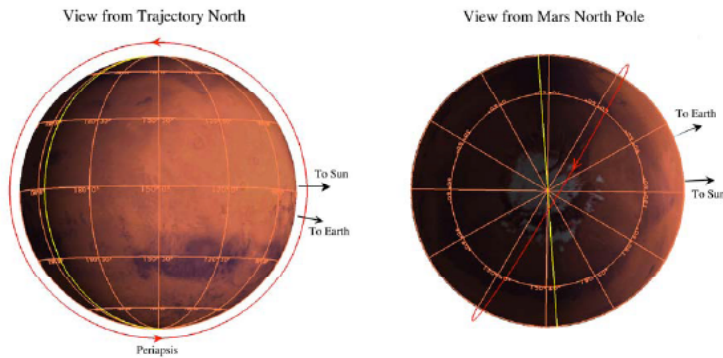


Figure 2. MRO primary science orbit (PSO)

The MRO mission architecture must address all of the objectives established for the project. The primary mission objective consisted of placing a science orbiter into Mars orbit to perform remote sensing investigations that will characterize the surface, subsurface and atmosphere of the planet and that will also identify potential landing sites for future missions. The MRO Primary Science Orbit (PSO) is a 255 x 320 km near-polar orbit, with the periapsis frozen over the South Pole, as shown in Fig. 2. This orbit is sun-synchronous with an ascending node orientation that provides

a Local Mean Solar Time (LMST) of 3:02 pm. A more detailed listing of the MRO scientific objectives can be found in Table 1.

To achieve these objectives, the MRO mission must accomplish six distinct mission phases: launch, cruise including trajectory correction maneuvers and instrument calibrations, Mars orbit insertion (MOI), aerobraking to lower the capture orbit into a low polar science orbit, a two-year primary science phase (PSP), and a two-year relay phase to support future landed assets on Mars. The data return for MRO will surpass 34 Tbits, more than all previous Mars missions combined. An overview of the MRO mission and operations is described in reference 1.

The challenge of system engineering, in the broadest sense, is to manage these extensive and diverse mission objectives into a single mission architecture that can be integrated, tested, launched, and operated. The system engineering team addressed these challenges across the various elements of mission, flight and ground system elements: mission design, end-to-end information system, spacecraft system and subsystems, science instruments and engineering payloads, operations processes, operations teams, and ground data system. As with most deep space missions, the architecture brings together a combination of proven designs from previous missions and new components that have never been flown. Some existing designs must be expanded to meet increased demands, and others interfaced into the architecture in new ways. This paper will discuss of some of the system engineering challenges that faced MRO, and their impact on the MRO flight and ground system designs.

Table 1. Mission objectives

• Characterize Mars' seasonal cycles and diurnal variations of water, dust, and carbon dioxide.
• Characterize Mars' global atmospheric structure, transport, and surface changes.
• Search sites for evidence of aqueous and/or hydrothermal activity.
• Observe and characterize the detailed stratigraphy, geologic structure, and composition of Mars surface features.
• Probe the near-surface Martian crust to detect subsurface structure, including layering and potential reservoirs of water and/or water ice.
• Characterize the Martian gravity field in greater detail relative to previous
• Mars missions to improve knowledge of the Martian crust and lithosphere and potentially of atmospheric mass variation.
• Identify and characterize numerous globally distributed landing sites with a high potential for scientific discovery by future missions.
• Provide navigation and data relay support services to future MEP missions
• Demonstrate Optical Navigation techniques for high precision delivery of future landed missions.
• Perform an operational demonstration of high data rate Ka-band.

II. Driving Requirements

Below is a summary of key driving requirements applicable to the system engineering challenges highlighted in this paper.

Science Operations: operate the orbiter and all six science instruments in the primary science orbit in targeting, survey and mapping modes, over the one Mars year of the primary science phase; conduct the gravity and

accelerometer investigations.

Data Volume: return, over the one-Mars-year primary science phase, a total science data volume return of 26 Tbits or more for full mission success. The minimum success criterion for data volume return is 15 Tbits or more.

Ephemeris Accuracy: predict the orbiter position and velocity relative to Mars to within 1.5 km in the downtrack direction, 0.05 km in the cross-track direction, and 0.04 km in the radial direction.

Targeting: spacecraft can be pointed anywhere in a ± 30 -degree cone about the nadir direction in PSO.

Simultaneous Operations of Instruments: support simultaneous operations of, and data acquisition by, all science instruments.

Science Planning: create a target acquisition selection process resulting in conflict free targeting priorities.

Product Telemetry: design and implement an application of the CCSDS File Delivery Protocol (CFDP).

Data Accountability: provide capability to identify, account for, transport, and reconstruct telemetry products produced by the science investigations and the ELECTRA relay payload.

Data Completion: provide to the spacecraft operators and science investigators at least 95% of data acquired.

Data Rates: support science downlink data rates up to 6 Mbit/sec.

Spacecraft Data Storage: provide onboard storage for 100 Gbits of science and science related data for the payload.

Retransmission: provide the capability for ground commanded retransmission of engineering and science data.

III. MRO Mission Operations Overview

The MRO project is managed by NASA's Jet Propulsion Laboratory (JPL) to provide oversight of the mission operations. This oversight includes integration of the flight system, its payloads, and the ground system used to operate it during its Mars mission. Systems engineering is embedded in all these elements, working together as part of teams located in multiple locations. Lockheed Martin Space Systems Company (LMSSC) in Denver, Colorado, is MRO's key system contractor. In addition to providing the MRO spacecraft itself, LMSSC continues to support MRO mission operations from Denver through all mission phases. The Principal Investigators (PI) and Team Leads (TL), each lead their own Science Operation Team (SOT), supporting science operations from remote locations across the United States and internationally (Rome, Italy) as well. This group comes together under the leadership of the project scientist and deputy project scientist to form the Project Science Group (PSG), working together strategically to review the best ways to meet project and instrument science objectives, and advise the project on key matters. Table 2 provides more detailed information about MRO's complement of science instrument payloads, their PIs and TLs, and team locations.

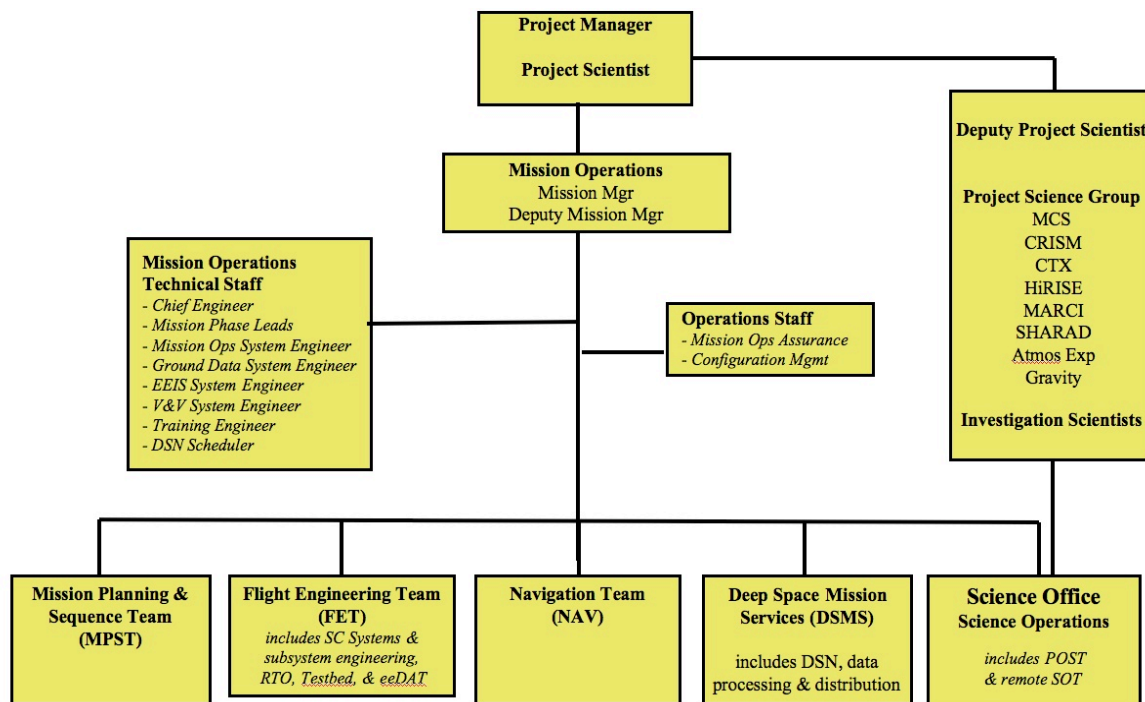


Figure 3. MRO flight operations team structure

Table 2. MRO science instrument descriptions

<i>Instrument</i>	<i>Type</i>	<i>Measurement Objectives</i>	<i>Science Goals</i>	<i>Attributes</i>
CRISM	High-resolution Imaging Spectrometer	• Hyper-spectral Image Cubes • 514 spectral bands, • 0.4 - 4 μm, 7 nm resolution • 20 m/pixel, 11 km swath	Regional & Local Surface Composition; Morphology	1. Moderately-High Spectral & Spatial Resolution 2. Targeted Observing & Global Survey 3. Very High Data Rate
CTX	Mono-chromatic Context Camera	• Panchromatic Images • 6 m/pixel; 30 km swath • Context Imaging for HiRISE/CRISM	Regional Stratigraphy; Morphology	1. High Resolution with Coverage 2. Targeted Observing & Regional Survey 3. High Data Rate
HiRISE	High-resolution Camera (0.5 m aperture)	• Color Images, • Stereo by Site Revisit • 0.3 m/pixel, 6 km swath (red) • 1.2 km swath (3 colors)	Stratigraphy; Geologic Processes; Morphology	1. Very High Resolution 2. Targeted Imaging 3. Very High Data Rate
MARCI	Wide-angle Color Imager	• Atmospheric cloud & haze, O_3, surface albedo • 7 bands: 0.28 - 0.8 μm	Global Weather; Surface Change	1. Daily Global Mapping, 2. Continuous Dayside Operations 3. Moderate Data Rate
MCS	Atmospheric Sounder	• Temperature, H_2O, Dust • Polar Radiation Balance • 0-80km vertical coverage • Vertical Resolution $\sim 5\text{km}$	Atmospheric Structure; Transport; Polar Processes	1. Daily Global Limb & Nadir Sounding, 2. Continuous Operations 3. Low-Data Rate
SHARAD	Shallow Subsurface Radar	• Ground Penetrating Radar • Split band at 20MHz • 10 - 20 m vertical resolution • 1 km x 5 km horiz. resolution.	Regional Near-Surface Ground Structure	1. Shallow Sounding 2. Regional Profiling 3. High Data Rate
Atmospheric Structure	Spacecraft Accelerometers	• Density and Scale Heights from spacecraft acceleration	Upper Atmosphere Structure and Circulation	Conducted during Aerobraking Phase only
Gravity Science	Spacecraft Telecom	• Spacecraft Range and Rate • X-band 2-way Doppler/Range	Gravity Field, Mass Distribution	Higher Resolution Seasonal Mass Redistribution
CRISM: PI, Scott Murchie, Johns Hopkins University Applied Physics Lab (JHUAPL) CTX: TL, Michael Malin, Malin Space Science Systems (MSSS) HiRISE: PI, Alfred McEwen, University of Arizona MARCI: PI, Michael Malin, Malin Space Science Systems (MSSS) MCS: PI, Daniel J. McCleese, Jet Propulsion Lab (JPL) SHARAD: TL, Roberto Seu, University of Rome, Italy; DTL Roger Phillips, Washington University Atmospheric Structure: TL, Gerald Keating, George Washington University at NASA LaRC Gravity: TL, Maria Zuber, Massachusetts Institute of Technology				

MRO will perform mission operations utilizing personnel from JPL, LMSSC, and multiple science operations sites. Dedicated MRO team leads are identified for each of the five operations teams that report directly to the MRO Mission Manager. As shown in Fig. 3, some of these operations teams include members from several MRO institutions. In the upcoming examples of system engineering challenges, several new operational teams established for MRO will be highlighted in more detail. Together, these teams will implement the day-to-day uplink and downlink processes that keep the mission going. The uplink processes designed for MRO work together to coordinate the operations teams with each other to produce and radiate the various engineering and science commands and sequences necessary to meet mission objectives. The downlink processes similarly coordinate the operation teams to provide data acquisition, real-time monitoring, and data processing. Figure 4 shows the family of MRO operational processes and their interfaces. Several of these processes will be specifically highlighted later in discussion within examples of system engineering challenges faced by MRO.

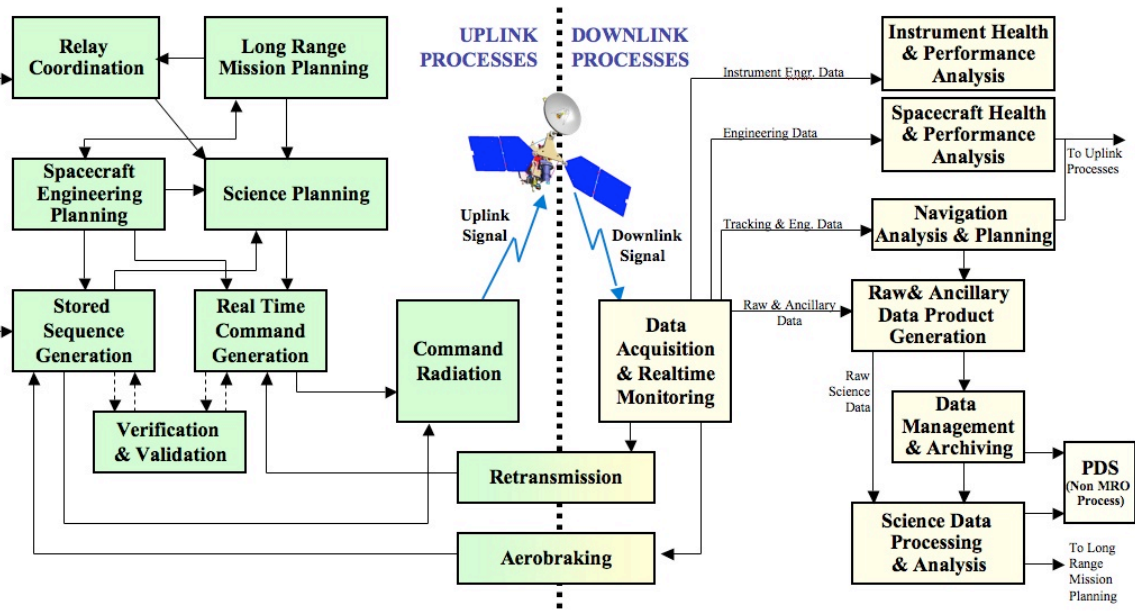


Figure 4. MRO mission operations processes

IV. Key System Engineering Challenges

A. Development of an Integrated Science Planning Process

MRO's complement of six science instruments creates a mission with a large set of exciting, but often conflicting, scientific objectives. Science teams will accomplish their objectives using three distinct observational modes: daily global mapping, regional surveys, and globally distributed targeting. Additional detail regarding each instrument's specific science can be found in Table 1. A conflict might arise, for example, when one instrument's off-nadir targeted observation interrupts another instrument's mapping or survey observations. The challenge of MRO's science planning process is to integrate each instrument's objectives and unique observation strategies into coordinated weekly timelines. This process must allow each team to select the observations they desire to further instrument science objectives, coordinate on joint observations, and de-conflict any overlapping observations in an equitable manner.

One of the spacecraft's most powerful capabilities is off-nadir targeting. Where previous Mars missions conducted mapping observations, instruments staring nadir, MRO has the capability of rolling ± 30 degrees off-nadir to conduct targeted observations. Science observations are implemented onboard the spacecraft via the Integrated Target List (ITL). An ITL file is uplinked to the spacecraft once a week, providing estimated time, orbit number, latitude and longitude for each science target. Using the ITL, along with regularly uplinked ephemeris and onboard command blocks, spacecraft flight software executes each targeted observation as specified. The ITL allows the science observations to be decoupled from the absolute-timed and relative-timed command sequences typically employed, like the MRO background sequences that handle spacecraft housekeeping functions and manage DSN contacts. The ITL, via the science planning process, is developed based on the spacecraft background sequence, but can be built and modified independently. The ITL allows for more accurate observation targeting by utilizing up-to-date spacecraft ephemeris, rather than estimating timing on the ground and using absolute-timed sequencing.

The solution to integrating all of these diverse objectives and observation strategies was to design a science planning process that divided the science planning process into two parts: interactive observation scheduling and non-interactive observation scheduling. Interactive Observations (IO) are observations that impact the operation of the spacecraft and/or another instrument such that SOTs must work together to schedule these observations. Examples of IOs are targeted observations, which require a spacecraft roll, and "high stability" observations, which require pausing the articulation of the spacecraft solar arrays and the MCS instrument. Non-Interactive Observations (NIO) are any instrument observations that do not affect the operation of the spacecraft or of any other instrument, thus nadir observations (no spacecraft roll).

In each science planning cycle, IO scheduling is completed first for the full 2-week cycle being planned, and next NIO scheduling is completed, one week of the two-week cycle at a time. To accomplish each segment of the science

planning process, the SOTs submit their inputs as a PTF (Payload Target File), which are merged together to produce IPTF (Integrated Payload Target File). At the end of each NIO planning week, an integrated, conflict-free schedule (“Final” IPTF) for all science observations is used to produce an ITL file for uplink to the spacecraft. A detailed description of the MRO science planning process can be found in reference 2.

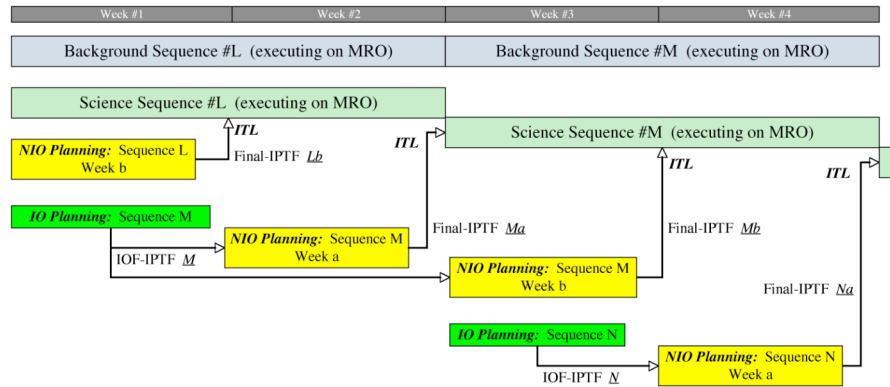


Figure 4. Science planning processes executing in parallel

One key attribute of the MRO science planning timeline is that it minimizes the overlap of individual planning cycles to the greatest extent possible. Ideally the ratio of planning to execution would be one-to-one or less, but due to the complexity of integrating IOs and NIOs into a single schedule, planning takes longer than execution. MRO’s science planning process requires 3.5 weeks to schedule each two weeks of execution, the last week of planning overlapping with the first week of execution. This means that multiple science planning cycles must be conducted in parallel, as shown in Fig. 4.

One of the most constraining factors in designing the science planning process was managing the on-board ephemeris files used by the ITL to conduct targeted observations. The requirement for spacecraft ground track error is 1.5 km in the downtrack direction and 0.05 km in the crosstrack direction. Detailed analysis conducted prior to launch revealed that onboard ephemeris files would need to be uplinked three times per week in order to ensure this accuracy.

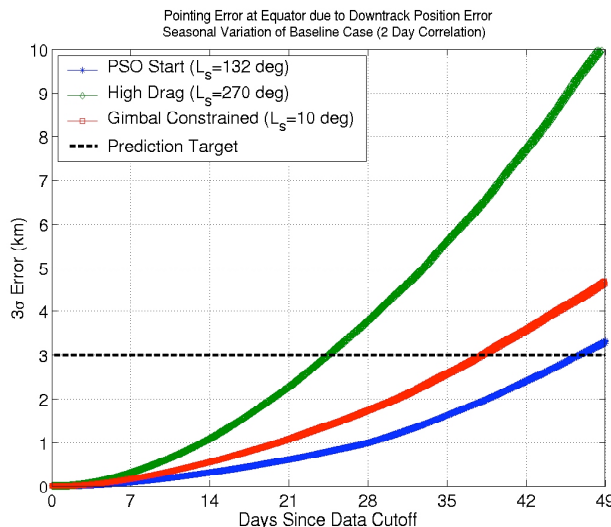


Figure 5. Ephemeris prediction accuracy over days since prediction

The accuracy of these predictions affects the science planning process in regards to orbit prediction, for both IO and NIO observations. For off-nadir IO’s, the rolls must be planned close enough to the eventual execution to fall within the ± 30 degree roll angle capability of the spacecraft. For NIO scheduling, the observations must be planned close enough to the eventual execution so that the desired target does not move outside the observations defined view. Figure 5 shows the accuracy of the ephemeris in days from the origination of the prediction. Once experience with actual primary science operations was gained in flight, the mission was able to reduce the number of ephemeris uploads per week from three to two for most of the Mars year.

Early in development, the project determined that a dedicated operations team to facilitate science planning, and produce the integrated observation schedules based on inputs from all science operations teams (SOTs)

was needed. MRO’s Payload Operations Support Team (POST), part of the science office shown in Fig. 3, consists of JPL support personnel for all science instruments to lead the science planning process, coordinating inputs (PTFs) from the SOTs to produce integrated science schedules (IPTFs) for the project. The POST provides the project a single point of contact for all science operations activities and issues, as well as acting as impartial arbiter in resolving operational conflicts between instruments teams.

In order to accomplish the ongoing science planning process, planning multiple science cycles in parallel, POST relies on a family of powerful science tools to automate observation scheduling. Two key tools stand out in making the MRO science planning possible within its defined timeline: Mars Targeting Tool (MTT) and Target Opportunity Scheduler (TOS). MTT can be run in a batch mode or an interactive (GUI) mode, which displays overlays of

observations on Mars MOLA (Mars Orbiter Laser Altimeter) maps, shown in Fig. 6. The POST routinely uses MTT to merge multiple PTFs into IPTFs, find and resolve conflicts between proposed observations, and generate a forecast of expected off-nadir roll angles for proposed observations. TOS is the core piece of software for generating a conflict-free schedule of Interactive Observations (IOs). This tool ingests prioritized lists of proposed observations (PTFs) from the SOTs and then performs a round-robin selection process, checking for conflicts or planning violations, scheduling as many observations as desired, and outputting the resulting IPTF.

Overall, the MRO science planning process has been incredibly successful. As of the release of this paper, over 16,000 science observations have been conducted since the beginning of the primary science phase. As a testament to the success of the system engineering process, the baseline science planning process established during the critical design review (February 2003) is in place today in operations. On a detailed level however, it is interesting to note that both the timeline and the tools have experienced almost constant adjustments between the start of primary science verification and validation (thread tests, rehearsals, and ORTs described in the next section) in October 2005, into the start of primary science phase in November 2006, and improvements are still in work today. This is a lingering result of balancing the diversity of science objectives and the commitment of the operations team to optimize the process in order to more fully achieve mission science objectives.

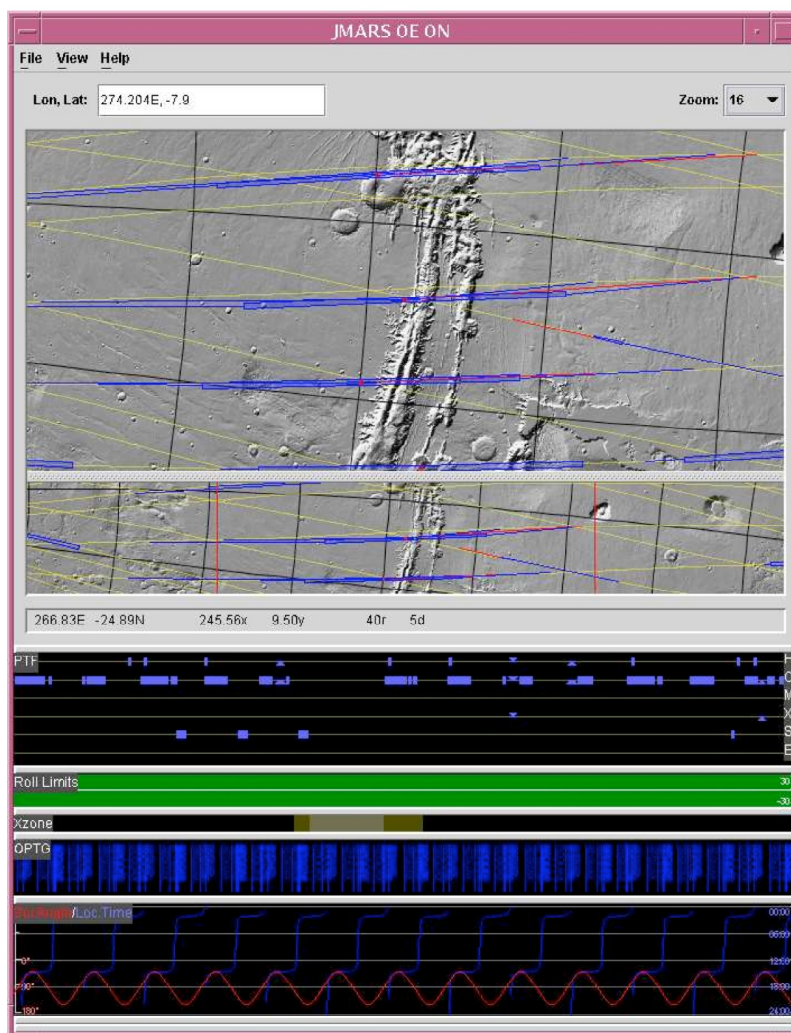


Figure 6. Mars Target Tool (MTT) screenshot showing science observations, blue overlay, on Mars MOLA maps

B. Implementation of End-to-End Information System

MRO data return is expected to well exceed 34 Tbits, made up science data from six instruments, engineering payloads, relay data from landed assets, and spacecraft engineering data. Managing this data, from the uplink of commands and sequences to the high rate downlink through the deep space network and including the data processing and distribution out to the remote science teams, is the job of the end-to-end information system (EEIS). Fig. 7 illustrates the complexity of MRO's EEIS system with its many interfaces in the flow of data, to and from the MRO spacecraft.

The product telemetry design for MRO is an application of the CCSDS File Delivery Protocol (CFDP) in an unacknowledged mode to identify, account for, transport, and reconstruct observation telemetry products produced by the MRO payloads. AOS (Advanced Orbiter System) frames transmitted are received by the DSN and forwarded to JPL via the DSN wide area network (WAN) where the MRO FDM (File Delivery Manager)/CFDP handler combination reassembles the data back into the raw CFDP data products originally created by each instruments observations. Products are not downlinked contiguously, but intermixed with other instrument data. Because the DSN WAN bandwidth can not support MRO's higher rates, MRO data will be stored at the DSN stations while

waiting to be forwarded to JPL (8 hour track of data is allowed up to 24 hours for delivery to JPL), thus frames for products received over multiple DSN stations may be received out of order by the FDM/CFDP handler processes that reassemble the original product. A set of CFDP timers manage each product transactions to ensure that it remains open an adequate amount of time to receive all data, including frames that may have to be retransmitted by the spacecraft if they are lost during their original transmission.

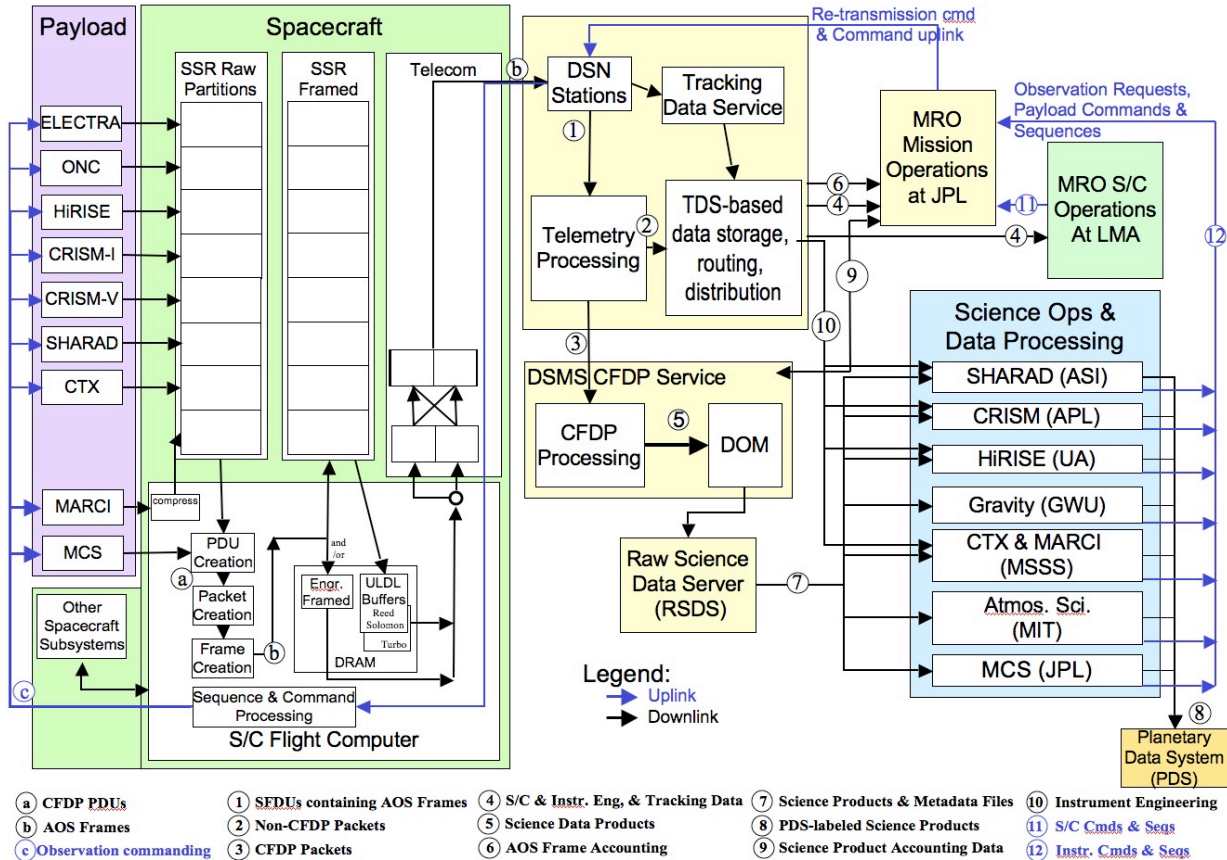


Figure 7. MRO's end-to-end information system

A significant challenge of this system was maintaining visibility of the status of each instruments data products as they travel through the EEIS. Unique product IDs indicate some status onboard as the data is passed from the instrument to the SSR and when the data is moved from raw to framed partitions, but this visibility disappears once the data has been framed on the spacecraft until it re-emerges on the ground during FDM/CFDP processing. Considering the large number of instrument data products to be produced daily, this makes tracking down missing data a much more complicated task, necessitating new tools that did not exist on previous missions. The MRO solution to this was the development of a new tool called TRUST (Tailored Report Utility for Science Product Tracking). TRUST monitors the movement of science product telemetry, from the science planning cycle, to spacecraft execution, through data downlink transmission. TRUST allows the science users to determine the status their data products within the EEIS using Internet accessible displays.

Early in project development, it was identified that MRO's high data rates and complex EEIS would require the special focus of a dedicated team to manage the high volume of data products to be received (between 2000-3000 per week). A new operations team would be necessary to manage the mission data, called the end-to-end Data Accountability Team (eeDAT). Part of the Flight Engineering Team (FET) shown in Fig. 3, eeDAT provides a focused point of contact support for the science operations teams to track and troubleshoot science data product delivery problems. This team also generates retransmission commands to recover lost data by downlinking it a second time from the spacecraft.

Several key lessons learned regarding the design, implementation and execution of the EEIS came to light 18 days after launch while conducting one of the first instrument checkouts in flight. MRO's high science data rates and

data volumes, that pushed the capabilities of the DSN and the entire EEIS, received the most attention during pre-launch testing. The engineering data flow was more typical of previous missions. During this instrument checkout, a large spacecraft backlog of engineering data stored on the SSR occurred due to high rate IMU (Inertial Measurement Unit) data collection that had not been terminated correctly by the operations team. When enabling the downlink of the SSR in order to receive the instrument calibration data products, the ground system unexpectedly had to deal with this high volume of engineering data that had accumulated and was assigned a higher priority “VC” (Virtual Channel) for downlink than the science data. Also unexpectedly, this overwhelmed the ground data processing capability, including a substantial lag, up to 30 minutes, of real-time data displays used to monitor the health and status of the spacecraft and payloads. This event exposed a design weakness and resulted in a series of actions that had to be implemented within the EEIS in order to be ready for upcoming mission phases. Analysis revealed engineering data backlogs would also occur during aerobraking phase of the mission, and the JPL data processing string could also get behind in processing, resulting in an unacceptable latency for the flight operations teams to execute time-critical aerobraking drag maneuvers. The project responded to this anomaly by: a) instituting limits on the spacecraft real-time data (VC 0) downlink to 100/338 kbps, b) updating the DSN to give MRO VC0 the highest priority when critical, c) upgrading JPL data processing hardware to increase processing capacity, d) upgrading the DSN WAN to add an additional T1 line of bandwidth, and e) conducting various levels of ground and in-flight testing. Although some of these issues were being worked prior to the flight anomaly, the importance of engineering data flows and interfaces had been underestimated in pre-launch analysis and testing.

C. Managing Onboard Data Storage and Retransmission

MRO utilizes a 160 Gbit Solid State Recorder (SSR) to store collected science and engineering data prior to transmitting this data back to Earth. MRO’s increased data rate capability, with rates up to 6 Mbit/sec, challenges the capacity of this space. Science operations teams must manage their data collection carefully in order to always take advantage of the available downlink, but without exceeding the limitations of the onboard storage capability. At the beginning the primary science phase (November 2006), the Earth-Mars distance was near its maximum, thus MRO began science data collection with its lowest data rates, less than 1.5 Mbit/sec. As the Earth-Mars distance decreases, the rates gradually increased to the maximum MRO rate of 6 Mbit/sec in August 2007. At this point in the mission the challenge is the greatest for the operations teams to design and continuously conduct valuable scientific observations to fill MRO’s downlink volume capability while not exceeding the SSR’s storage capacity.

MRO’s complement of six science instruments adds an additional complexity, in that each of these payloads collects data in a different manner, targeting observations that collect data in a bursty manner, versus mapping instruments that collect data in a more continuous regular manner. Additionally, each instrument collects data at different rates and produces data products in a wide variety of sizes ranging from a few kbits (MCS) to 24 Gbits (HiRISE). In order to accommodate MRO’s multiple payloads, the SSR’s memory is partitioned into multiple areas providing space for each payload. Each payload is allocated both a “raw” and a “framed” storage space within the SSR (shown as part of Fig. 7). The raw space is used for the storage of the raw science data written directly from the payload before it is AOS framed by spacecraft flight software. The SSR framed space is used for the storage of these AOS framed science data awaiting downlink.

The design challenge was to determine a way to provide operational independence to each instrument to manage their SSR storage space with minimal impact from the data collection of other instruments, and also ensure operational fairness in regards to the division of the downlink bandwidth amongst teams. The solution was to provide each instrument a dedicated hard partition for the storage of raw science data. Each instrument’s interface software will control the writing of data to its raw partition, and the spacecraft FSW is responsible for read-out of instrument data from the raw partition to the processing buffer. The framed data area of the SSR is shared rather than dedicated, configured as a single large hard partition for X-band framed data (Ka-band framed data partition was essentially zeroed out after the Ka-band exciter failure during aerobraking). This framed hard partition is subdivided into smaller soft partitions providing space for each instrument, which is managed by the spacecraft’s flight software. Both SSR hard partition sizes, and framed soft partition sizes, are configurable by uplinking new configuration files to the spacecraft. Determination of each instrument’s SSR allocations is negotiated and managed by MRO’s Project Science Group (PSG). Key factors considered when determining the division of SSR space between teams were the maximum sizes of each instrument’s data products and the anticipated rates of collection to meet project and instrument science objectives. Management of each payload’s raw and framed SSR partitions is the responsibility of the SOT for each instrument. So far the SSR has been reconfigured twice during the mission to improve efficiency of onboard data storage and improve the retransmission capability.

MRO’s retransmission capability is another factor to consider in managing the onboard storage of data. It is not possible to wait for the science data products to be produced on the ground to determine their completeness: by the

time these products are processed, any missing data would most likely be overwritten in the SSR. Instead, ground software detects frame gaps at the DSN, representing lost science and engineering data transmitted by the spacecraft. Retransmission commands sent by the ground operators can request retransmission of the lost frames, increasing the data quality for the mission, but without knowing exactly which data is missing. For the ground, these commands must be generated as quickly as possible: After collecting the frame gap data, the retransmission commands must still be built and radiated promptly to avoid overwriting the data onboard. One of the limiting factors in is the staffing of flight controllers, called “ACEs”, required to uplink commands. Since other spacecraft operations do not require this support most of the time, retransmission commanding opportunities are limited to two or three times per day. Retransmission can thus be improved by managing the science data collection onboard as well. By minimizing the backlog of science data awaiting downlink, the already transmitted data will be retained longer in the SSR, giving the ground more time to request retransmission.

MRO’s onboard data management design described above has proven itself an invaluable asset to the mission. It provides reliable data storage and downlink capability to multiple payloads, but it is complex. Each science team requires the capability to model their data flow onboard the spacecraft when conducting the science planning process for their observations in upcoming science cycles. This is accomplished by using a combination of tools available to the science teams. Although each team has developed tools that are optimized for their own instruments operations, the basic approach involves a simplified SSR model built into their own home-grown science planning software, followed by final checking by project-provided software called Data Tracker. SOT home-grown data volume checking involved ingesting a JPL-provided downlink data file (called “DLINK” file, a listing of data volumes per DSN contact) into a simple two-reservoir model of SSR raw and framed space for their single instrument. Data Tracker software, developed by LMSSC as the provider of the onboard data management design, performs high fidelity modeling of the spacecraft and multi-payload data system reflecting the most accurate prediction of flight.

The ground retransmission process also developed a sophisticated set of tools. Based on frame gaps detected at the DSN, retransmission request generation software automatically generates retransmission commands to be transmitted to the spacecraft. This, when paired with retransmission scheduler software, will produce a retransmission request when scheduled uplinked opportunities are available.

One lesson learned was that more attention needed to be dedicated to validating science data volume estimates and modeling. Primary science operations had already begun when an error in the DLINK file was discovered which resulted in overestimating the allowable science data volume available for downlink during some DSN contacts. This resulted in science teams using more SSR storage than anticipated and limited the success of retransmission.

Another lesson learned was the extent of how highly leveraged the retransmission process was to the science teams. Small data losses occur routinely due to a variety of reasons, such as space losses, DSN configuration problems, or spacecraft and commanding idiosyncrasies. These may not seem to have a significant impact from an overall data volume point of view ($< 2\%$), but even a small frame loss in a science product can cause data processing calibration issues that require manual attention. As a result of its importance to the science teams, additional effort was devoted to increase the effectiveness of retransmission. An example of this additional effort was making an update to the onboard SSR configuration to include a new soft partition in the framed space for retransmission. This retransmission soft partition was allocated 20% of the framed space, but unlike the science payload soft partitions, no data will ever be produced for it. This space became a buffer region, so that when the science teams were more fully utilizing onboard storage space, the flow of real instrument data from the raw partitions would be slowed down, increasing the time available to request retransmission of framed data before it was overwritten.

D. Design of Operational Readiness Tests

System engineering is a key player in the verification and validation of any system; where verification shows that we built the system right to meet the design requirements, and validation shows that we have built the right system for the mission. Developing a good project level test program is challenging; to bring all key flight and ground system elements together in the most flight-like manner possible, and to demonstrate mission operations as they will occur accurately, is no simple task. MRO relied on three main system test types to accomplish this task: End-to-End Information System Tests, Sequence Verification Tests, and Operational Readiness Tests. These are defined below:

1. End-to-End Information System Tests (EEIST): tests of end-to-end information flow including flight, ground, and DSN interfaces
2. Sequence Verification Tests (SVT): flight and ground system tests to validate planning, development, execution, and analysis of critical and typical mission scenarios, during nominal and off-nominal conditions.

3. **Operational Readiness Tests (ORT):** demonstrate readiness of mission operations teams and ground system to conduct flight operations including anomaly response, for critical and typical mission scenarios.

Each of these test types fulfill specific and sometimes unique objectives of verifying and validating the project system, and offered many challenges in order to meet the adage “test like you fly”. Developing flight-like data sets, use of spacecraft simulators, and establishing good initial test conditions are key examples. Using flight-like command sequences and involving the actual operation teams that will fly the mission are other good practices in developing tests.

One of most challenging test designs to develop for MRO were the set of primary science Operational Readiness Tests. Even though this phase was not considered a critical event like launch or Mars orbit insertion (MOI), from a test perspective, it offered a number of unique qualities that made this a particularly challenging scenario to develop:

- The primary science phase activities are a complex overlay of sequence and science planning, commanding and execution, and data downlink including data flow out to remote science team sites.
- For a two-week science execution and downlink period, 3.5 weeks of planning are required with one week overlapping. This would result in a 4.5-week test to complete in a flight like manner.
- Key components to the complexity of PSP operations are the data rates and resulting data volumes available. The higher rates are more taxing on the team operationally, however it seemed unrealistic to validate new processes and tools with inexperienced teams with the most stressful case the first time.
- Unlike the launch and MOI testing, which could be completed within one day using the MRO’s high fidelity Orbiter Test Bed (OTB), the PSP sequence would require two weeks. Prior to these ORTs, for the longest OTB test ran for nine days before crashing.
- MRO PSP processes, tools, and interfaces offered the least heritage from previous Mars missions. There would likely be a large numbers of issues identified during initial testing that would have to fixed and retested.

In response to these challenges, a series of test activities were defined to validate PSP processes leading up to a final high rate ORT. First, several science planning thread tests were performed to validate the overall process interfaces and to checkout tools. These tests were distinguished by being focused primarily on the science planning process and tools; other processes were not exercised and strict timelines were not followed. If extra time was required for a particular step of the process to workout an issue, it was allowed. Second, two rehearsals (one low rate and one high rate) were performed to

integrate the science planning process with other uplink processes. A stricter timeline was enforced for these rehearsals. Finally, two ORTs (again, one low rate and one high rate) integrating both uplink and downlink processes into the most flight like demonstration of PSP operations. Fig. 8 shows how the combination of thread tests, rehearsals, and ORTs fit over the planned data volume timeline for PSP.

One of the key features of this PSP test plan was the way in which the rehearsals and ORTs fit together strategically to limit the actual test timelines to 2.5 weeks each. Each rehearsal produced spacecraft sequences, science planning products and instrument command that would be loaded onto the OTB during the ORTs. The low rate rehearsal outputs were used for the low rate ORT, and likewise for the high rate test cases. Since the ORTs must demonstrate both uplink and downlink processes simultaneously, these flight-like products were essential to operate OTB at the start of the test so that not only would the spacecraft portion produce engineering data in a flight like manner, the instruments engineering models attached to the OTB would produce flight-like science data products that would flow through the JPL portion of the EEIS and out to the remote science team sites and exercise initial portion each teams data processing pipeline.

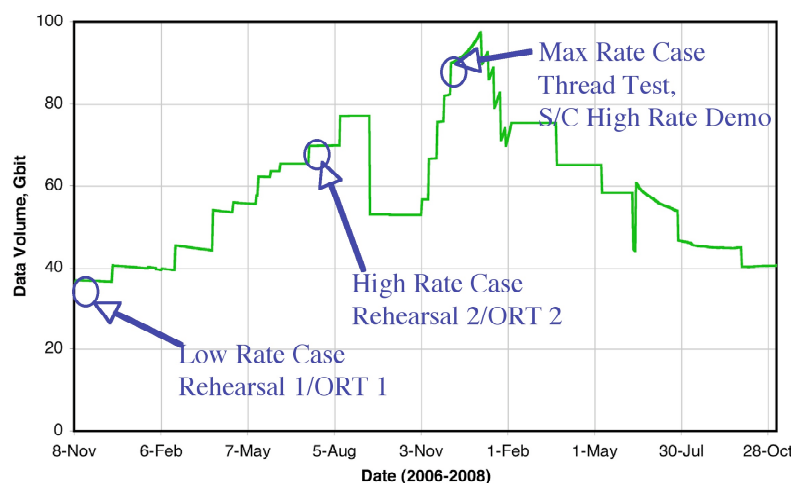


Figure 8. Primary Science Test Cases

A key challenge during the this period was the parallel execution of MRO flight operations through cruise, MOI, and aerobraking mission phases in conjunction final primary science development and testing. Each mission operations team had to allocate some team members to focus on primary science, while others focused on flight operations. Rehearsals and ORTs were scheduled carefully around key events, standing down completely for MOI. It took 11 months from the start of the post launch thread test to the completion of ORT, but this was time well spent. At the end of each test, actions and lessons learned were captured and resolved in preparation for the next test. These improvements included updates to ground tools, processes, and command sequences. When MRO began its PSP operations in November 2006, although challenging, operations were well served by the thorough testing. Overall the MRO operations teams were able to adhere to the PSP timeline from the very beginning, and no significant issues emerged.

V. Conclusion

MRO is continuing its two-year primary science phase, begun on November 7, 2006. In early July 2007, MRO met its requirement to collect 15 Tbits of data after only eight months of operations. Based on current performance the MRO mission is expected to return 70 Tbits of data by the end of the primary science phase in 2008. The MRO system has met the challenge of its driving requirements.

Systems engineering challenges were overcome utilizing a combination by creative designs built into MRO's flight and ground systems. Key components of these designs were the establishment of a strong operations team organization, robust operational processes, and the development strategic ground tools.

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Acknowledgments

The research and development described here was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration. A large number of people from the MRO Project, Lockheed Martin Space Systems Company, and MRO science teams contributed to the development of the designs and operational processes described within this paper. I would like to express my appreciation to all of them.